Climate Change: Evidence, Impacts, and Practical Responses

The basic science

Earth’s climate is influenced by the balance between incoming energy from the Sun and energy leaving the planet. Certain gases in the atmosphere absorb and re-emit some outgoing infrared energy, warming the surface and lower atmosphere. This natural greenhouse effect makes Earth habitable. Human activities have increased the concentration of greenhouse gases, strengthening the effect and changing the climate.

Carbon dioxide is released mainly by burning coal, oil, and natural gas, as well as by some industrial processes and land-use changes. Methane comes from sources including fossil fuel systems, agriculture, and waste. Nitrous oxide is associated with fertilizer use and other activities. These gases differ in how long they remain in the atmosphere and how strongly they trap heat, but all contribute to warming.

Evidence of a changing climate

Multiple independent observations show that the planet is warming. Measurements from weather stations, ships, satellites, and ocean instruments indicate rising global average temperatures. Glaciers are shrinking in many regions, sea level is rising, and the ocean is accumulating heat. These patterns are consistent with the expected effects of increased greenhouse gases.

Climate change does not mean that every location becomes hotter every day. Natural variability still affects local weather, and some regions may experience colder-than-average periods. The long-term trend is assessed using many years of observations, not a single season. Scientists also compare observed changes with climate models and physical understanding to estimate the influence of different factors.

Impacts on people and ecosystems

A warmer atmosphere can increase the likelihood or intensity of some heat extremes. Higher temperatures can raise the risk of heat illness, affect outdoor work, and increase demand for cooling. Changes in rainfall patterns can contribute to drought in some places and heavier downpours in others. The consequences depend on local geography, infrastructure, and preparedness.

Sea-level rise increases coastal flood risk and can worsen erosion and saltwater intrusion. Ecosystems may shift as temperature and seasonal conditions change; species that cannot move or adapt may face greater stress. Agriculture can be affected by heat, water availability, pests, and changing growing seasons. These effects are uneven, and communities with fewer resources often have less capacity to prepare and recover.

Mitigation: reducing future warming

Mitigation means reducing greenhouse-gas emissions or increasing the removal of carbon dioxide from the atmosphere. Cleaner electricity, energy efficiency, electrified transport, and lower-emission industrial processes can reduce emissions. Protecting forests and restoring ecosystems can preserve carbon stores, although land-based solutions must be designed carefully and cannot replace rapid emissions cuts.

Cities can support mitigation through compact planning, reliable public transport, safe walking and cycling, efficient buildings, and clean energy. Businesses can measure emissions across operations and supply chains, then prioritize reductions that are credible and measurable. Individuals can influence demand through travel, energy use, and purchasing choices, while also supporting broader policy and infrastructure changes.

Adaptation and resilience

Adaptation reduces harm from climate impacts that are already occurring or expected. Examples include heat action plans, shade and cooling centers, flood-resilient infrastructure, water conservation, and early-warning systems. Nature-based approaches—such as restoring wetlands or planting appropriate urban trees—can provide cooling, absorb water, and support biodiversity when planned for local conditions.

Good adaptation starts with identifying who and what is exposed, which hazards are likely, and what resources are available. Plans should include vulnerable groups and be updated as conditions change. Adaptation has limits: some impacts become increasingly difficult to manage as warming grows, which is why reducing emissions and preparing for impacts need to proceed together.

Making progress

Climate responses involve governments, businesses, communities, and individuals. Public policy can set standards, fund infrastructure, and encourage cleaner technologies. Research and innovation can improve energy systems, materials, agriculture, and monitoring. Communities can identify local priorities and make plans that reflect their needs.

Progress is easier to evaluate when goals are specific and transparent. Useful measures include emissions trends, energy efficiency, resilience investments, and whether vulnerable populations are better protected. Climate change is a large and complex challenge, but its causes and many responses are well understood. Sustained action can reduce future risks and create additional benefits such as cleaner air, healthier cities, and more reliable energy systems.